
Cleaning and exploring data

A recipe for analysis-ready data

September training - session 1

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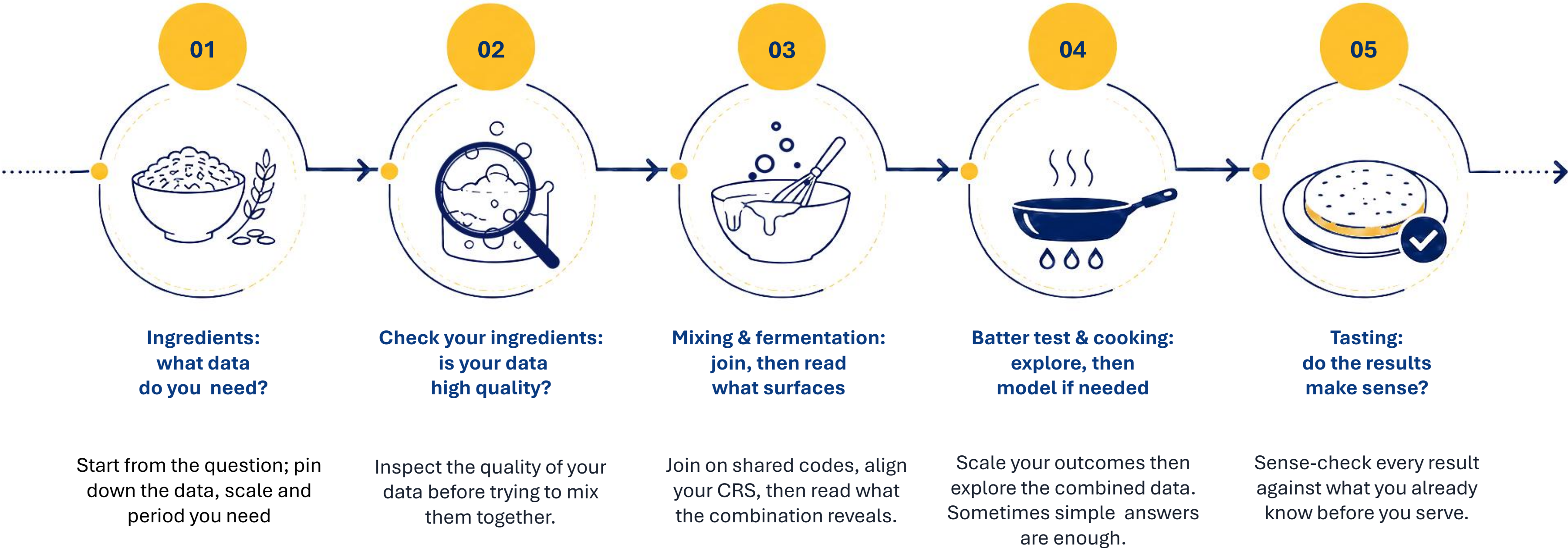
Recap: a recipe for analysis-ready data



Understanding data through the lens of making Injera



Recap: the recipe for analysis-ready data



Agenda

00 Background and the question

02 Check your ingredients: is your data high quality?

04 Batter test & cooking: explore, then model if needed

06 Conclusion

01 Ingredients: what data do you need?

03 Mixing & fermentation: join, then read what surfaces

05 Tasting, and writing the recipe down

SECTION 00

Background



Deciding what data you actually need to answer your question



Background

- This session is an extension of the July session and focuses on a practical application of the recipe
- The data used throughout this session are **simulated** and intended purely for pedagogical purposes
- You work in the nutrition coordination office, and you receive the following email:

From: Nutrition Coordinator, Oromia Regional Health Bureau

To: Analysis team

Subject: Rainfall and child nutrition

Rainfall has been poor across parts of the region for the past two seasons. I need to know whether that is showing up in child nutrition.

We have routine growth monitoring records from health posts across 14 zones, and rainfall estimates for the same months. There is no budget or time for a survey.

Two things, please: **is there an association between rainfall and malnutrition and which zones should we focus on?**

I need something I can take to the planning meeting **in two weeks.**

Background

- The context for the session is as follows:
 - You are received a request from the Oromia Regional Health Bureau
 - Oromia has been heavily impacted by a lack of rainfall (drought)
 - You need understand if this influences malnutrition in children and which zones are most impacted
- You consider the following question:

Question 1

Which zones in Oromia have the highest malnutrition?

Question 2

Is rainfall associated with child malnutrition in Oromia?

Poor rains → crop & livestock losses → household food insecurity → malnourishment

SECTION 01

Ingredients: what data do you need?



Deciding what data you actually need to answer your question



Questions to consider when determining what data is needed

Given our questions:

1. Which zones in Oromia have the highest malnutrition?
2. Is rainfall associated with child malnutrition in Oromia?

Consider:

1. What data is needed to address this question? (Think about the ideal world scenario)
2. What data is available in Ethiopia?
3. If the data is not available, what data is available in other countries with similar settings or is there a proxy you can use?
4. What data can be feasibly collected in the future to refine addressing question?

Task

In your group, take 5 minutes to discuss question 1

Data availability

You enquire about is available in Ethiopia based on the ideal dataset you would like, and are told that the following datasets are available:

Dataset	Format	One row per	Period	Columns
Screening register	.csv	person × visit	2022–2024	record_id, person_id, woreda_name, health_post, visit_date, entry_date, sex, dob, age_value, age_unit, weight_kg, height_cm, muac_mm
Rainfall estimates	.csv	zone × month	2019–2024	zone_pcode, date, n_pixels, rain_mm, rain_lta_mm, rain_anom_pct
Population projections	.csv	woreda × year	2019–2024	woreda_pcode, woreda_name, year, total_pop, pop_under2, pop_under5
Woreda boundaries	.shp	woreda	2024	wor_pcode, wor_name, zone_pcode, zone_name, reg_pcode, reg_name, geometry
Health post locations	.csv	health post	-	health_post, hp_name, lon, lat

Check your ingredients: is your data high quality?

You will learn to:

1. Use a variable summaries to understand your data and find problems
2. Assess a single dataset against six standard data quality dimensions
3. Recognise that the dimensions are not independent
4. How to fix common data issues



Data quality inspection

Task

Inspect the screening register dataset and assess each data quality dimension below. Address any issues you find.

1 Completeness

Data are complete when no field the *intended use* depends on is missing.

2 Accuracy

Data are accurate when they correctly capture the true event they represent.

3 Validity

Data are valid when they conform to the expected type, format, range, and categories.

4 Consistency

Data are consistent when values agree within a record and across datasets.

5 Uniqueness

Data are unique when there are no duplicates.

6 Timeliness

Data are available rapidly enough for the intended use.

Data quality results

1. Completeness

- MUAC is 80% missing at one health post and 58% at another, and complete everywhere else
- Missing weights and heights are MCAR; missing MUAC tracks the health post $\Rightarrow$ MAR

2. Accuracy

- 35 weights recorded in grams, 26 heights in metres, and 31 ages in months labelled as years

3. Validity

- Sex arrives in 10 codings, and the codes 1 and 2 pass silently
- Woreda names arrive with inconsistent case and spacing, 205 raw names cleaning to the expected 57 (56 sampled plus Unknown)

4. Consistency

- Thirty children carry a conflicting sex and 17 a conflicting date of birth

5. Uniqueness

- 74 rows are exact copies
- 273 child-days carry more than one record (the same child on the same day with a new record ID but a different weight)

6. Timeliness

- Records in one woreda reach the system a median of 97 days after the visit, against about a week everywhere else
- The final month holds 219 records against a typical 500-570, so it is dropped as a partial month

Data quality takeaways

- Good data entry is **extremely** important and will reduce the work required at this stage
- Some checks depend on one another, e.g. validity and consistency
- Different dimensions may be more important depending on the task
- You may need to apply judgement at this stage
- There are multiple frameworks, this is just an example of one

Food for thought:

Can we create a consolidated data scoring system in Ethiopia?



SECTION 03

Mixing & fermentation: join, then read what surfaces

In this Section, you will learn to:

1. Describe what a shapefile contains, and explain why every operation carries the geometry with it
2. Check and convert the coordinate reference system before joining layers
3. Count how many records matched a spatial join, and map those that did not
4. Different ways to join records to spatial data (by name or co-ordinates)
5. Combine data to a higher geographic level



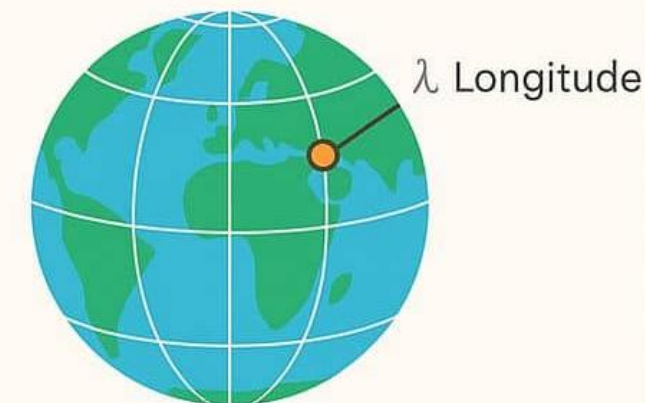
A few reminders

- A CRS is how you turn the Earth's curved surface into coordinates
- You can join tabular data using a shared column
- You can data to join to a polygon by:
 - Name
 - Coordinates (point-in-polygon)
 - Overlapping polygons
- Joins can cause missing data when keys do not match; find and fix the mismatch instead of filling in the missing values

What to do:

- Match the CRS across every layer before you join
- Pick something reliable to join on
- Check your join for missing or duplicate rows, and anything out of place

Geographic Coordinate System (GCS)



- Based on Earth's spherical surface
- Units: Degrees (°)

Projected Coordinate System (PCS)



- Based on a 2D flat surface (map)
- Units: Meters or Feet

Working with shapefiles

Task

Join the screening register, population projections, estimated rainfall to a shape file.

1 What's in a shapefile

Open the boundary file and read its header before the rows. Note the geometry column.

2 Plotting shapefiles

Plotting shape files. Note the difference between filtering and zooming.

3 Coordinate systems

Creating an sf object from tabular data and joining files with different coordinate systems.

4 Joining: point-in-polygon

Different ways to perform a point-in-polygon join and diagnosing issues.

5 Joining: name

Joining by raw names compared to an identifier.

6 Aggregating up

Combine wordas into zones and join the rainfall data.

7 Tying it all together

Finally, join everything together into an analysis ready dataset.

Throughout we will be using the [sf package](#) which is an R package used for working with shape files

Working with shapefiles results

1. What's in a shapefile

- The header tells you the geometry type, the bounding box and the CRS
- Every row carries a geometry defining its shape

2. Plotting shape files

- There are different ways to display shape file data; filtering vs zooming give two very different views

3. Coordinate systems

- `st_join` gives an error when the CRS does not match

4. Joining: point-in-polygon

- There are different options for how to treat boundary points
- Four health posts landed in the Mediterranean – lon/lat entered the wrong way round; swapping them recovers all four

5. Joining: name

- Four rows are lost when joining by name, but no rows are lost when using an identifier code
- Joining population multiplies the rows by the number of years and repeats the geometry

6. Aggregating up

- Rainfall lives at zone level $\Rightarrow$ dissolve when you want the shape, aggregate and join when you want the number

7. Tying it all together

- Notice the assumption you make about rainfall when joining to the woreda level nutrition and population data

Working with shapefiles takeaways

- A shapefile has a geometry which is carried through every operations $\Rightarrow$ working with shape files can be resource intensive, only use the geometry when you need it
- You cannot disaggregate data without making assumptions $\Rightarrow$ you're usually limited by your coarsest dataset
- Joining by a common name is the least reliable method when joining spatial data
- Joining by point-in-polygon is a better option, but beware of points on polygon boundaries
- Objects need to have the same CRS system if you're making a spatial join
- Joining can remove rows or duplicate them; you need to count rows before and after each join

SECTION 04

Batter test & cooking: explore, then model if needed

In this Section, you will learn to:

1. Create an outcome variable that can be compared across zones
2. Use exploratory analysis to inspect the outcome and answer policy questions



Investigating the data and answering the policy questions

Task 3

Turn weights into WHO z-scores, then explore the data across space, across time, and against rainfall, with the aim of answering our two questions. You will produce combined_dataset - the analysis dataset - and an answer to each question

Question 1

Which zones in Oromia have the highest malnutrition?

Question 2

Is rainfall associated with child malnutrition in Oromia?

1 From a weight to an outcome

Create WHO weight-for-age z-scores, so that we can compare weights across ages.

2 Look at the outcome

Plot the z-scores against the -2 and -3 thresholds, then split the same distribution by zone.

3 Mapping the underweight

Map the number of children underweight and the proportion of those screened who are underweight. Note the difference.

4 Coverage

Use the under-5 population as a denominator, then rank the zones by proportion underweight alongside coverage.

5 Over time

Look at the proportion underweight and the rainfall anomaly month by month.

6 The relationship

Relate the proportion underweight to rainfall three to six months earlier.

Investigating the data and answering the policy questions

Question 1

Which zones in Oromia have the highest malnutrition?

Answer:

Ilu Aba Bora, East Hararge, Guji and Borena have the highest proportion of screened children underweight, all at or above 20%. That is almost double the lowest zone, West Wellega, at 11%.

Question 2

Is rainfall associated with child malnutrition in Oromia?

Answer:

Yes, as rainfall decreases the proportion of underweight children increases. Borena, Guji and East Hararge stand out as most at risk - they combine a high proportion of underweight children with series that track rainfall closely.

And what to be careful about when you present it, be upfront about your limitations

1 Fewer rainfall observations

Every child in a zone shares one rainfall value: 14 exposure trajectories, not thousands, and only 50-65 zone-months per zone.

2 Nothing at woreda level

Measured on zone-months, with sampled woredas standing in for whole zones, so differences between woredas are hidden.

3 Likely an overestimate

Screening is not random. Caregivers may bring the children who already look sick, pushing every zone's proportion above true prevalence.

4 Screened ≠ all children

Coverage is under 1% in every zone, so none of these figures estimates prevalence in the population.

5 No effect size

We show only that the two move together, not how much rainfall effects the outcome.

6 The mechanism is assumed

Rainfall is considered on its own: no harvests, food prices, conflict or distance to care. Two ends of an assumed chain.

What to consider if a model is needed

- If you do need a model to answer your question keep the following in mind:

Before you model

- Document every cleaning decision
- Freeze an analysis-ready dataset you can point back to

While you model

- Document any data problems the model uncovers; a wild coefficient can be a data error, not a finding.

After you model

- Fix any data issues in the raw dataset, version it again, and rerun from scratch

Food for thought:

In injera: once the batter hits the mitad, you can't add water back. In analysis: once you model, prep is locked in.



SECTION 05

Tasting: do the results make sense?



Sense-checking your results before you trust and share them



Does it taste right? Sense-check outputs

- Regardless of the analysis, before you share your results, check them against what is known

Four potential checks:

1. Against known **benchmarks**: do your rates sit near published estimates (national or in the literature)?
E.g. an order-of-magnitude gap may mean a units or denominator error
2. Against **theory**: do effects point the way theory expects, and are they plausibly sized?
E.g. a protective effect for a known risk factor may be a red flag
3. Against **totals**: do modelled counts reconcile with raw totals you trust?
E.g. a mismatch may signal a join error or double-counting
4. Sensitivity to **your choices**: would a different reasonable choice flip the conclusion?
E.g. a conclusion that flips under a different cleaning choice means that is, not the data, is driving your result

Food for thought:

A result you can't explain to a domain expert isn't ready to serve. Interrogate it before you trust it.



SECTION 06

Conclusion



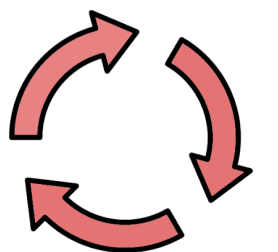
Reproducibility: write the recipe down

- It's important keep track of what you've done so that your analysis is reproducible and so that you can reuse the dataset for other analyses if necessary.



Four things to keep in mind:

1. Script it, don't hand-edit: make changes in your code, rather than manual edits. What you have done to the raw data needs to be reproducible.
2. Notice throughout: quality isn't a final step. Check your dataset and address issues at every step.
3. Save your ersho: keep the cleaned dataset, the code, and the metadata (calendars, boundaries, CRS) and a written record of what you've done and any assumptions you've made. This is your starter for your next analysis.
4. Remember the feedback loop: your analysis may reveal a data gap or a better way to frame your question - go back and address it, rather than forcing your analysis to fit what you already have.



What we covered and what comes next

What we covered:

- **Check your ingredients** Six quality dimensions on one register. They depend on each other, so diagnose everything first and fix once.
- **Mix and ferment** A shapefile is a table that carries shapes. Match the CRS, join on codes, count what matched and map what did not.
- **Batter test** Create an outcome that can answer your questions, then explore it. Try to answer your question with out a model first.

What comes next:

- **A model?** We described the relationship. A model would say how much it explains, and whether it holds once age, season and zone are accounted for.
- **Refine the question?** We only had rainfall. Harvests, food prices, conflict and distance to care are all missing from the file. We also answered the question at a zone level with a sample of woredas, could we go more granular?



Making Good Data Is Like Making Injera

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It's important to follow the correct steps and be careful along the way. You don't fix bad injera - or bad data - at the end. You learn to notice problems throughout the process.

Thank you

